

# Joseph AO

## Software Engineer

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*First Class Software Engineering graduate with a portfolio of systems-level C++ projects — including a Redis-style in-memory server built from scratch. Experienced in analytical, evidence-driven problem-solving through two years in high-scale FinTech. Passionate about systems programming and building robust, high-quality software.*

## SKILLS

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### Languages

C++, C, Bash, Python, JavaScript

### Systems & Tooling

Event-driven I/O, non-blocking sockets, poll(), thread pools, pipelining, /proc filesystem, Linux, Git, GDB, g++/clang++, Makefile, VS Code

### Data Structures & Algorithms

Hash tables (incremental rehashing), AVL trees, min-heaps, doubly linked lists, sorted sets, sorting, searching,  $O(\log n)$  range queries, TTL expiry scheduling, graph traversal

### Methodologies

- Agile (Prince2), Project Planning, Risk Management, cross-functional team collaboration

## WORK EXPERIENCE

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### Monzo - Fraud Investigator

Jan 2024 - Jan 2026

London(Remote)

- Analysed complex transaction datasets and built evidence timelines across high-volume fraud cases at one of the UK's largest digital banks, applying rigorous, systematic reasoning to ambiguous data.
- As Learning & Development Champion, developed structured workshops and mentoring programs that addressed gaps in coding skills and communication, resulting in a 15% reduction in project delivery times.
- Applied FCA and GDPR compliance standards in all decision-making processes, resulting in a 20% increase in data processing efficiency by overhauling internal tooling workflows.

### TSYS EMEA - Fraud Analyst

July 2023 - Jan 2024

Milton Keynes, MK7 8LE

- Analysed card fraud and disputes using open-source intelligence and third-party data; resolved over 300 cardholder claims in compliance with Visa/Mastercard association guidelines

### Industrial Training Fund(NYSC) - Executive Administrative Assistant

Dec 2018 - Dec 2020

- Designed technical assessments and automated scoring tools; managed 500+ application records with 95% placement success rate and 20% reduction in recruitment cycle time

### AI Model Trainer (RLHF) - Outlier

Nov 2024 - current

Remote

- Evaluated Python code and a set of 100+ mathematical problems to enhance LLM output quality, contributing to reinforcement learning from human feedback pipelines.

## PROJECTS

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### Redis-Style In-Memory Server - C++ · Systems Programming

- Built a fully event-driven server in C++ from scratch; non-blocking sockets, single-threaded poll() event loop, request pipelining, and explicit per-connection read/write state management
- Custom hash table with incremental rehashing (two-table system) to eliminate resize latency spikes; AVL-tree-backed sorted sets for  $O(\log n)$  range queries
- Min-heap TTL expiry with per-entry index references for  $O(\log n)$  deletion; idle connection tracking via doubly linked list
- Thread pool offloads expensive cleanup (large ZSet teardown) to avoid blocking the I/O loop; custom length-prefixed binary protocol supporting key-value, sorted set, and TTL commands

### Linux System Monitor - Bash · Linux

- Reads CPU deltas directly from /proc/stat with real-time RAM, disk, network and top-5 process display; demonstrates practical Linux internals — /proc filesystem, awk/grep processing, iproute2 networking tools

### Bank Account System - C++ · OOP

- Encapsulated Account class, header/implementation separation, deposit/withdrawal/overdraft logic, transaction history via std::vector, and file persistence with fstream

### EA Sports Software Engineering Simulation - C++ · Forage

- Implemented Strategy Design Pattern across four abstract base classes in C++; applied composition-over-inheritance via unique\_ptr, virtual destructors, and RAII throughout a 324-line header

## EDUCATION

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### University of Bedfordshire - Bachelor's of Science, Software Engineering

July 2023, First Class

Luton

### Aptech Computer Education - Advanced Diploma, Software Engineering

Distinction

### University of Jos - Bachelor's of Science, Physics Education

Jos, Plateau

## CERTIFICATIONS

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- Python Developer Masters Program — Edureka (in progress) — Python Programming, Data Science, AI, PySpark, Django
- EA Sports Software Engineering Job Simulation (Forage) — C++, Strategy Design Pattern, UML, debugging
- Programming for Everybody: Getting Started with Python — University of Michigan / Coursera · 98.34%